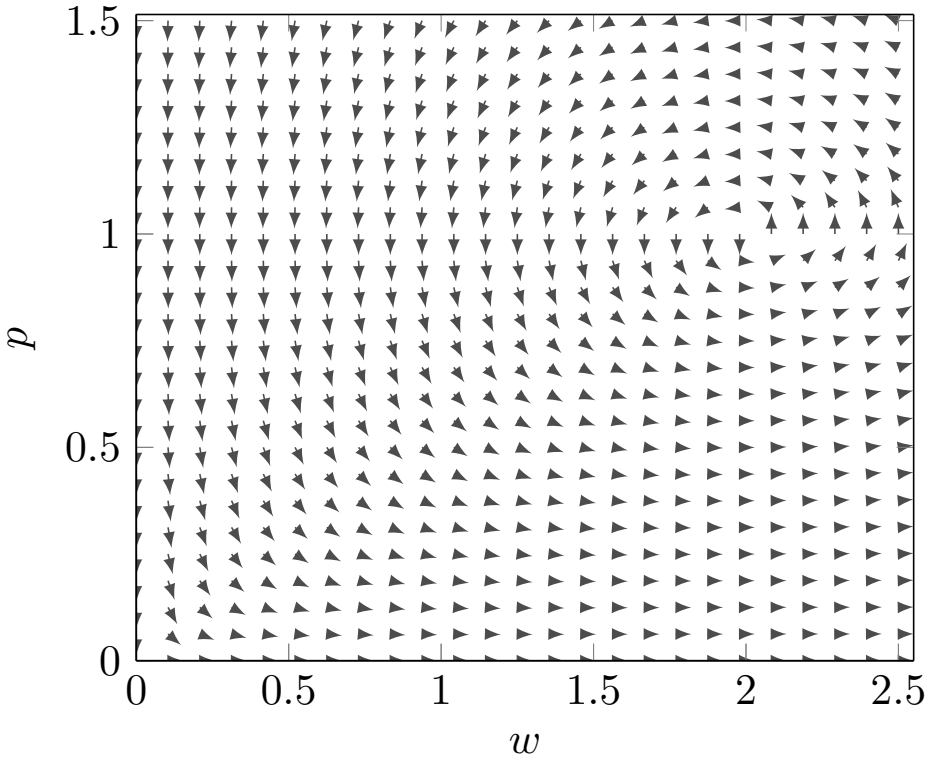
17 1. The population of the threatened brush-tailed rock-wallaby is being monitored in real-time in a conservation park in the presence of a predators population. Suppose the populations are measured in thousands over time t and considered as continuous variables, $w(t)$ for wallabies and $p(t)$ for the predator.

The populations of the wallabies and predator are related by the system of ODEs

$$\begin{aligned} \frac{d\,w}{dt} &= w - w\,p \\ \text{and} \quad \frac{d\,p}{dt} &= -p + \frac{1}{2}\,w\,p. \end{aligned} \tag{1}$$

The terms involving the product $w\,p$ create interactions between the two populations.

The direction field (i.e., vector field with normalised vector lengths) in phase space for (1) is shown below. It is restricted to the relevant quadrant for the problem, $w, p \geq 0$.



(a) Briefly describe in words what would happen to the two populations without the interactions (i.e., without the $w\,p$-terms)?

Solution: The wallaby population would increase and the predator population would decrease (both exponentially).

(b) Explain why the interaction terms in (1) imply that p is the predator population and w is the prey population.

Solution: The signs of the interaction terms means that they reduce the population of the wallabies and increases the predator population.

(c) Is the system (1) linear or nonlinear? Explain your answer.

Solution: Nonlinear because it involves products of unknowns in the interaction terms $(w\,p)$.

(d) You are given that $(w, p) = (0, 0)$ is a critical point of (1). Find the one other critical point $(w, p) = (w_*, p_*)$.

Solution: The critical point requires

$$w - w\,p = 0 \quad \text{and} \quad -p + \frac{1}{2}\,w\,p = 0.$$

As we discount the case $w = p = 0$, the remaining solution is

$$p = 1 \quad \text{and} \quad w = 2$$

so that $(w_*, p_*) = (2, 1)$.

(e) Close to the critical point $(w, p) = (0, 0)$, (1) is approximately the linear system

$$\begin{aligned} \frac{d\,w}{dt} &= w \\ \frac{d\,p}{dt} &= -p. \end{aligned} \tag{2}$$

Express (2) as a vector-valued, first-order, linear ODE of the form

$$\boldsymbol{y}' = \mathbf{A}\,\boldsymbol{y} \quad \text{where} \quad \boldsymbol{y}(t) = \begin{pmatrix} w(t) \\ p(t) \end{pmatrix}, \tag{3}$$

i.e., find the matrix $\mathbf{A}$.

Solution: The matrix $\mathbf{A}$ is

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

(f) Find the eigenvalues and eigenvectors of $\mathbf{A}$ and, hence, write down the general solution for $\boldsymbol{y}(t)$.

Solution: The eigenvalues and eigenvectors are

$$\lambda_1 = 1 \quad \text{and} \quad \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

and

$$\lambda_2 = -1 \quad \text{and} \quad \boldsymbol{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hence, the general solution is

$$\boldsymbol{y}(t) = C_1\,\mathrm{e}^{t}\,\boldsymbol{v}_1 + C_2\,\mathrm{e}^{-t}\,\boldsymbol{v}_2$$

(g) Close to the critical point $(w, p) = (w_*, p_*)$, we let $w(t) = w_* + W(t)$ and $p(t) = p_* + P(t)$, where W and P are small and obey the linear system

$$\begin{aligned} \frac{dW}{dt} &= -w_* P \\ \frac{dP}{dt} &= \frac{p_*}{2} W. \end{aligned} \tag{4}$$

Express (4) as a vector-valued, first-order, linear ODE of the form

$$\boldsymbol{z}' = \mathbf{B}\,\boldsymbol{z} \quad \text{where} \quad \boldsymbol{z}(t) = \begin{pmatrix} W(t) \\ P(t) \end{pmatrix}, \tag{5}$$

i.e., find the matrix $\mathbf{B}$.

Solution: The matrix $\mathbf{B}$ is

$$\mathbf{B} = \begin{pmatrix} 0 & -2 \\ \frac{1}{2} & 0 \end{pmatrix}$$

(h) Find the eigenvalues and eigenvectors of $\mathbf{B}$. Hence, write down the general solution for $\boldsymbol{z}(t)$ in terms of real-valued quantities.

Solution: The eigenvalues and eigenvectors are

$$\mu_1 = \mathrm{i} \quad \text{and} \quad \boldsymbol{u}_1 = \begin{pmatrix} 2 \\ -\mathrm{i} \end{pmatrix}$$

and

$$\mu_2 = \overline{\mu_1} = -\mathrm{i} \quad \text{and} \quad \boldsymbol{u}_2 = \overline{\boldsymbol{u}_1} = \begin{pmatrix} 2 \\ \mathrm{i} \end{pmatrix}$$

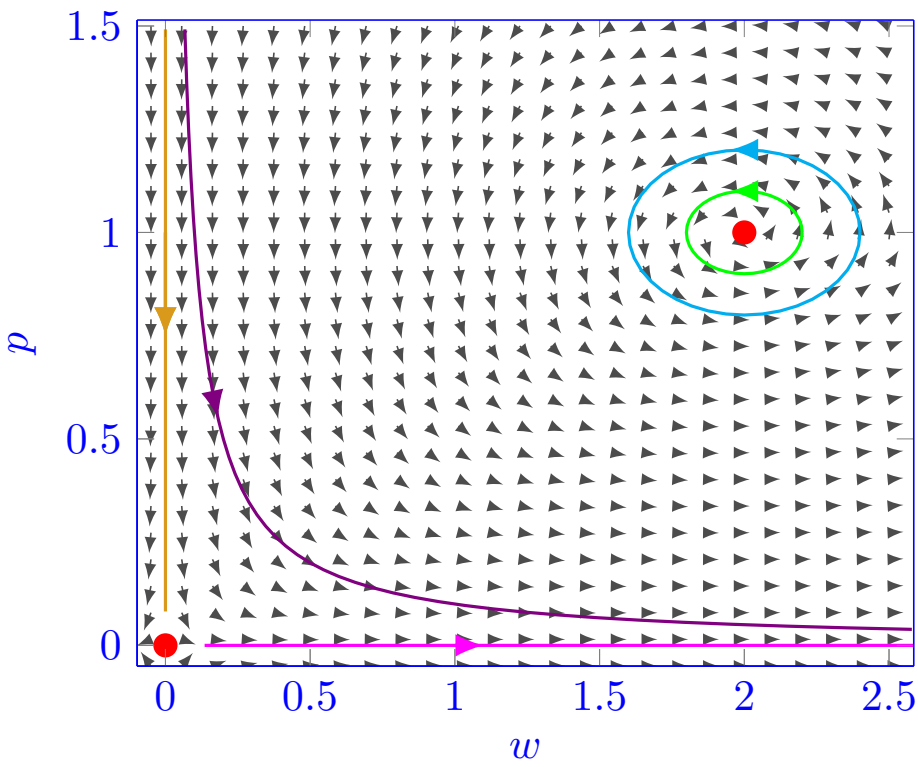
Hence, the general solution is

$$\begin{aligned} \boldsymbol{y}(t) &= C\,\mathrm{e}^{\mathrm{i}t}\,\boldsymbol{u}_1 + \overline{C\,\mathrm{e}^{-\mathrm{i}t}\,\boldsymbol{u}_2} \\ &= 2\,\mathrm{Re}\{C\,\mathrm{e}^{\mathrm{i}t}\,\boldsymbol{u}_1\} \\ &= A \begin{pmatrix} 2\cos(t) \\ \sin(t) \end{pmatrix} + B \begin{pmatrix} -2\sin(t) \\ \cos(t) \end{pmatrix}. \end{aligned}$$

(i) Sketch the phase portrait, including the critical points, representative trajectories around the critical points, and directions of the trajectories.

You can sketch on top of the direction field given above, or create your own phase space to sketch on.

Solution:



- The critical points
- Straight-line trajectories along axes
- Trajectories around the critical point at the origin
- Trajectories around the critical point not at the origin
- Directions

(j) What are the stabilities of the two critical points?

Solution: The critical point at the origin in unstable and the critical point not at the origin is stable (although not asymptotically stable).

(k) In terms of the real-world problem, briefly explain whether you think it is preferable to start with initial populations close to the critical point $(w, p) = (0, 0)$ or (w_*, p_*)?

Solution:

A possible answer is that initial populations close to (w_*, p_*) is preferable, as then both populations survive and do not grow unboundedly.

$\boxed{8}$ 2. Consider the following second-order linear ODE

$$2\,t^2\,\frac{\mathrm{d}^2u}{\mathrm{d}t^2} + 3\,t\frac{\mathrm{d}u}{\mathrm{d}t} - u = g(t) \quad \text{for} \quad t > 0. \tag{6}$$

(a) Verify that $u_1(t) = t^{-1}$ is a complementary solution of equation (6).

Solution: The required derivatives are

$$u_1 = t^{-1} \quad \Rightarrow \quad u_1' = -t^{-2} \quad \Rightarrow \quad u_1'' = 2\,t^{-3}.$$

Substituting into the LHS of (6) gives

$$\begin{aligned} 2\,t^2\,\frac{\mathrm{d}^2u_1}{\mathrm{d}t^2} + 3\,t\frac{\mathrm{d}u_1}{\mathrm{d}t} - u_1 &= 2\,t^2\,2\,t^{-3} + 3\,t(-t^{-2}) - t^{-1} \\ &= 4\,t^{-1} - 3\,t^{-1} - t^{-1} = 0, \end{aligned}$$

as required.

(b) Use Abel's theorem to find a second linearly independent complementary solution, $u_2(t)$. Check your answer.

Solution:

Abel's theorem gives

$$\begin{aligned} \mathrm{wr}[u_1, u_2](t) &= C\,\mathrm{e}^{-\int 3\,t\,/\,2t^2\ \mathrm{d}t} \\ &= C\,\mathrm{e}^{-(3\,/\,2)\ln(t)} \\ &= C\,t^{-3\,/\,2} \end{aligned}$$

where C is a constant.

Thus

$$\begin{aligned} u_1\,u_2' - u_1'\,u_2 &= C\,t^{-3\,/\,2} \\ \Rightarrow \quad t^{-1}\,u_2' + t^{-2}\,u_2 &= C\,t^{-3\,/\,2} \\ \Rightarrow \quad u_2' + t^{-1}\,u_2 &= C\,t^{-1\,/\,2} \\ \Rightarrow \quad [I\,u_2]' &= C\,I\,t^{-1\,/\,2} \end{aligned}$$

where I is the integrating factor

$$I = \mathrm{e}^{\int t^{-1}\ \mathrm{d}t} = t.$$

Hence,

$$t\,u_2 = C\int t^{1\,/\,2}\ \mathrm{d}t = \hat{C}\,t^{3\,/\,2} \quad \Rightarrow \quad u_2 = \hat{C}\,t^{1\,/\,2} \quad \text{where} \quad \hat{C} = \frac{2\,C}{3}.$$

Thus, a second linearly independent complementary solution is $u_2(t) = t^{1\,/\,2}$

Checking

$$u_2 = t^{1\,/\,2} \quad \Rightarrow \quad u_2' = \frac{1}{2}\,t^{-1\,/\,2} \quad \Rightarrow \quad u_2'' = \frac{-1}{4}\,t^{-3\,/\,2}$$

so that

$$2\,t^2\,\frac{\mathrm{d}^2 u_1}{\mathrm{d}t^2} + 3\,t\frac{\mathrm{d}u_1}{\mathrm{d}t} - u_1 = 2\,t^2\left(\frac{-1}{4}\,t^{-3/2}\right) + 3\,t\left(\frac{1}{2}\,t^{-1/2}\right) - t^{1/2}$$
$$= -\frac{1}{2}\,t^{1/2} + \frac{3}{2}\,t^{1/2} - t^{1/2} = 0,$$

as required.

(c) Use reduction of order to find the general solution of (6) when $g = 0$. Briefly comment on the consistency of the solution with your findings from the previous question.

Solution: The reduction of order formula from lectures is

$$y(t) = v\int\frac{1}{I}\left[\int\frac{I\,g}{v}\,\mathrm{d}t\right]\,\mathrm{d}t \quad \text{where} \quad I(t) = v^2\,\mathrm{e}^{\int f_1\mathrm{d}t}.$$

Here,

$$f_1(t) = \frac{3}{2\,t}, \quad g(t) = 0/2\,t^2 = 0 \quad \text{and} \quad u_1(t) = t^{-1},$$

so that the reduction of order formula gives

$$u(t) = t^{-1}\int\frac{1}{I}\left[\int\frac{I\times 0}{t^{-1}}\,\mathrm{d}t\right]\,\mathrm{d}t$$

where

$$I(t) = t^{-2}\,\mathrm{e}^{\int f_1\mathrm{d}t} = t^{-2}\,\mathrm{e}^{\int 3/(2\,t)\,\mathrm{d}t} = t^{-2}\,t^{3/2} = t^{-1/2}.$$

Therefore,

$$\begin{aligned} u(t) &= t^{-1}\int t^{1/2}\left[\int 0\,\mathrm{d}t\right]\,\mathrm{d}t \\ &= C_1\,t^{-1}\int t^{1/2}\,\mathrm{d}t \\ &= C_1\,t^{-1}\left[\frac{2}{3}\,t^{3/2} + C_2\right] \\ &= D_1\,t^{-1} + D_2\,t^{1/2}, \end{aligned}$$

where D_1 and D_2 are constants.

Thus, the general solution to the homogeneous problem gives the solution space spanned by u_1 and u_2, as expected.

(d) Use reduction of order to find the general solution of (6) when $g = t$. Check your answer.

Solution: Applying the reduction of order formula with $g = t/(2\,t^2) = 1/(2\,t)$ (noting

that the notation g is not consistent with that used in this question) , we have

$$\begin{aligned}
u(t) &= t^{-1} \int t^{1/2} \left[\int \frac{t^{-1/2} \times (1/2)\, t^{-1}}{t^{-1}} \, \mathrm{d}t \right] \, \mathrm{d}t \\
&= \frac{1}{2}\, t^{-1} \int t^{1/2} \left[\int t^{-1/2} \, \mathrm{d}t \right] \, \mathrm{d}t \\
&= \frac{1}{2}\, t^{-1} \int t^{1/2} \left[C_1 + 2\, t^{1/2} \right] \, \mathrm{d}t \\
&= \frac{1}{2}\, t^{-1} \int \left[C_1\, t^{1/2} + 2\, t \right] \, \mathrm{d}t \\
&= \frac{1}{2}\, t^{-1} \left[C_2 + \frac{3}{2}\, C_1\, t^{3/2} + t^2 \right] \, \mathrm{d}t \\
&= \frac{1}{2}\, C_2\, t^{-1} + \frac{3}{4}\, C_1\, t^{1/2} + \frac{1}{2}\, t.
\end{aligned}$$

The $\frac{1}{2}\, C_2\, t^{-1} + \frac{3}{4}\, C_1\, t^{1/2}$ part is the solution space for the homogeneous problem, as in part (c). Let the remaining term be

$$u_{\mathrm{p}} = \frac{t}{2} \quad \Rightarrow \quad u_{\mathrm{p}}' = \frac{1}{2} \quad \Rightarrow \quad u_{\mathrm{p}}'' = 0.$$

Thus

$$\begin{aligned}
2\, t^2\, u_{\mathrm{p}}'' + 3\, t\, u_{\mathrm{p}}' - u_{\mathrm{p}} &= 2\, t^2 \times 0 + 3\, t \times \frac{1}{2} - \frac{t}{2} \\
&= t,
\end{aligned}$$

as required.

$\boxed{7}$ 3. Consider the following system of first-order linear ODEs:

$$\begin{aligned} \dot{x} &= 2\,y - t, \\ \dot{y} &= 3\,x - y. \end{aligned} \tag{7}$$

The dots indicate derivatives with respect to time, t. Convert this system into a single vector-valued first-order linear ODE. Find the general solution by matrix diagonalization. Check your solution.

Solution: As a vector-valued, first-order ODE, (7) is

$$\underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x \\ y \end{pmatrix}}_{=\dot{\boldsymbol{x}}} = \underbrace{\begin{pmatrix} 0 & 2 \\ 3 & -1 \end{pmatrix}}_{=\mathbf{A}} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{=\boldsymbol{x}} + \underbrace{\begin{pmatrix} -t \\ 0 \end{pmatrix}}_{=\boldsymbol{b}}.$$

The matrix $\mathbf{A}$ has eigenvalues $\lambda_1 = -3$ and $\lambda_2 = 2$, with corresponding eigenvectors

$$\boldsymbol{v}_1 = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \quad \text{and} \quad \boldsymbol{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Thus, $\mathbf{A}$ is diagonalised as

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix}^{-1}.$$

The diagonalised system is

$$\dot{\boldsymbol{y}} = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix} \boldsymbol{y} + \boldsymbol{c} \quad \text{where} \quad \boldsymbol{y} = \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix}^{-1} \boldsymbol{x}$$

and

$$\begin{aligned} \boldsymbol{c} &= \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix}^{-1} \boldsymbol{b} \\ &= \frac{1}{5}\begin{pmatrix} 1 & -1 \\ 3 & 2 \end{pmatrix} \boldsymbol{b} \\ &= \frac{1}{5}\begin{pmatrix} -t \\ -3\,t \end{pmatrix}. \end{aligned}$$

Solving, e.g., with an integrating factor, we have

$$\boldsymbol{y} = \begin{pmatrix} C_1\,\mathrm{e}^{-3\,t} \\ C_2\,\mathrm{e}^{2\,t} \end{pmatrix} + \begin{pmatrix} \frac{1}{45} - \frac{1}{15}t \\ \frac{3}{20} + \frac{3}{10}\,t \end{pmatrix}$$

Thus,

$$\begin{aligned} \boldsymbol{x} &= \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix} \boldsymbol{y} \\ &= C_1 \begin{pmatrix} 2 \\ -3 \end{pmatrix} \mathrm{e}^{-3\,t} + C_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \mathrm{e}^{2\,t} + \boldsymbol{x}_\mathrm{p} \end{aligned}$$

where $\boldsymbol{x}_\mathrm{p}$ is the particular solution

$$\begin{aligned}\boldsymbol{x}_\mathrm{p} &= \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{45} - \frac{1}{15} t \\ \frac{3}{20} + \frac{3}{10}\, t \end{pmatrix} \\ &= \begin{pmatrix} \frac{7}{36} + \frac{1}{6}\, t \\ \frac{1}{12} + \frac{1}{2}\, t \end{pmatrix}.\end{aligned}$$

Checking the complementary solution,

$$\begin{aligned} \boldsymbol{x}_\mathrm{c} &= C_1 \begin{pmatrix} 2 \\ -3 \end{pmatrix} \mathrm{e}^{-3\,t} + C_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \mathrm{e}^{2\,t} \\ \Rightarrow \quad \boldsymbol{x}_\mathrm{c}' &= -3\, C_1 \begin{pmatrix} 2 \\ -3 \end{pmatrix} \mathrm{e}^{-3\,t} + 2\, C_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \mathrm{e}^{2\,t} \\ &= \mathbf{A}\, \boldsymbol{x}_\mathrm{c} \end{aligned}$$

as required.

Checking the particular solution,

$$\begin{aligned} \boldsymbol{x}_\mathrm{p}' &= \begin{pmatrix} \frac{1}{6} \\ \frac{1}{2} \end{pmatrix} \\ \mathbf{A}\, \boldsymbol{x}_\mathrm{p} &= \begin{pmatrix} \frac{1}{6} + t \\ \frac{1}{2} \end{pmatrix} \\ \Rightarrow \quad \boldsymbol{x}_\mathrm{p}' - \mathbf{A}\, \boldsymbol{x}_\mathrm{p} &= \begin{pmatrix} \frac{1}{6} - \frac{1}{6} - t \\ \frac{1}{2} - \frac{1}{2} \end{pmatrix} = \begin{pmatrix} -t \\ 0 \end{pmatrix}, \end{aligned}$$

as required.

$\boxed{8}$ 4. Consider the ODE

$$9\,t\,(1-t)\,y''(t) + 6\,(2-3\,t)y'(t) - 2\,y(t) = 0. \tag{8}$$

(a) Does the existence theorem for solutions of ODEs from lectures guarantee a continuous solution at $t = 0$? Justify your answer.

Solution: In standard form, (8) is

$$y''(t) + \frac{6\,(2-3\,t)}{9\,t\,(1-t)}\,y'(t) - \frac{2}{9\,t\,(1-t)}\,y(t) = 0.$$

The coefficients are not continuous at $t = 0$ and hence the existence theorem does not guarantee a continuous solution.

(b) Apply a power series solutions to the ODE (8) in the form

$$y(t) = \sum_{n=0}^{\infty} a_n\,t^n, \tag{9}$$

and find a recurrence relationship for the coefficients.

Solution:

Substituting (9) into the ODE gives

$$\sum_{n=0}^{\infty} 9\,t\,(1-t)\,n\,(n-1)\,a_n\,t^{n-2} + \sum_{n=0}^{\infty} 6\,(2-3\,t)\,n\,a_n\,t^{n-1} - \sum_{n=0}^{\infty} 2\,a_n\,t^n = 0$$

$$\Rightarrow \quad -\sum_{n=0}^{\infty} 9\,n\,(n-1)\,a_n\,t^n + \sum_{n=0}^{\infty} 9\,n\,(n-1)\,a_n\,t^{n-1}$$

$$+\sum_{n=0}^{\infty} 12\,n\,a_n\,t^{n-1} - \sum_{n=0}^{\infty} 18\,n\,a_n\,t^n - \sum_{n=0}^{\infty} 2\,a_n\,t^n = 0$$

$$\Rightarrow \quad -\sum_{m=0}^{\infty} 9\,m\,(m-1)\,a_m\,t^m + \sum_{m=-1}^{\infty} 9\,(m+1)\,m\,a_{m+1}\,t^m$$

$$+\sum_{m=-1}^{\infty} 12\,(m+1)\,a_{m+1}\,t^m - \sum_{m=0}^{\infty} 18\,m\,a_m\,t^m - \sum_{m=0}^{\infty} 2\,a_m\,t^m = 0$$

$$\Rightarrow \quad \sum_{m=0}^{\infty} \Big\{ -9\,m\,(m-1)\,a_m + 9\,(m+1)\,m\,a_{m+1}$$

$$+12\,(m+1)\,a_{m+1} - 18\,m\,a_m - 2\,a_m \Big\}\,t^m = 0$$

$$\Rightarrow \quad \sum_{m=0}^{\infty} \Big\{ 3\,(m+1)\,(3\,m+4)\,a_{m+1} - (3\,m+1)\,(3\,m+2)\,a_m \Big\}\,t^m = 0.$$

Thus, we deduce the recursion relationship

$$a_{m+1} = \frac{(3\,m+1)\,(3\,m+2)}{3\,(m+1)\,(3\,m+4)}\,a_m \quad \text{for} \quad m = 0, 1, 2, \ldots$$

(c) Comment on the number of linearly independent solutions defined by the recurrence relationship.

Solution: There is only one free coefficient in the recurrence relationship (a_0) and, hence, it only gives one linearly independent solution.

(d) Determine the radius of convergence of the power series (9).

Note that finding the radius of convergence **does not** require you to check the case $L = 1$.

Solution: Use the ratio test

$$\lim_{n\to\infty}\left|\frac{a_{n+1}\,t^{n+1}}{a_n\,t^n}\right| = \lim_{n\to\infty}\left|\frac{(3\,n+1)\,(3\,n+2)}{3\,(n+1)\,(3\,n+4)}t\right|$$

$$= \lim_{n\to\infty} t.$$

Thus, the radius of convergence is one.

(e) Trial a solution of the form t^λ to find the second solution to the ODE.

Solution: Substituting $y = t^\lambda$ into (9) gives

$$\begin{aligned} 9\,t\,(1-t)\,\lambda\,(\lambda-1)\,t^{\lambda-2} + 6\,(2-3\,t)\,\lambda\,t^{\lambda-1} - 2\,t^\lambda &= 0 \\ \Rightarrow \quad -9\,\lambda\,(\lambda-1)\,t^\lambda + 9\,\lambda\,(\lambda-1)\,t^{\lambda-1} + 12\,\lambda\,t^{\lambda-1} - 18\,\lambda\,t^\lambda - 2\,t^\lambda &= 0 \\ \Rightarrow \quad 3\,\lambda\,(3\,\lambda+1)\,t^{\lambda-1} - (9\,\lambda^2 + 9\,\lambda + 2)\,t^\lambda &= 0. \end{aligned}$$

Thus, we require

$$\lambda\,(3\,\lambda+1) = 0 \quad \text{and} \quad (9\,\lambda^2 + 9\,\lambda + 2) = 0,$$

which has solution $\lambda = -1\,/\,3$. We have found the second solution $y = t^{-1\,/\,3}$.

(f) Are your solutions consistent with your answer to (a)?

Solution: Yes, as $y = t^{-1\,/\,3}$ is not continuous at $t = 0$.